Reasons for optimism

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An attempt to rectify the rather confusing discussion of optimism in ESL using pedantic notation.

I. INTRODUCTION

In statistical learning, we fit models to data. It's a bad idea to use the training error to assess model performance due to overfitting; on the other hand, it can be hard to guess how well the model will do on unseen data unless we have a lot of data, or a very strong prior about the model. *Optimism* is an estimable quantity somewhere in between, and leads to various model selection criteria, including the AIC.

Let's be more precise. Suppose we have a noisy function from $\mathcal{X}$ to $\mathcal{Y}$; this means that data is generated according to

$$\mathbb{P}(X, Y) = \mathbb{P}(X)\mathbb{P}(Y|X), \tag{1}$$

i.e. Y is always conditional on X. A *training set* is a collection of N independent samples from this distribution,

$$\mathcal{T} = \{(x_1, y_1), \ldots, (x_N, y_N)\}. \tag{2}$$

We use $\mathbf{x} = (x_i)$ and $\mathbf{y} = (y_i)$ for the observed vectors of inputs and outputs, and sometimes write $\mathcal{T} = (\mathbf{x}, \mathbf{y})$.

A *model* is a family of functions $\mathcal{F} \subseteq \mathcal{Y}^{\mathcal{X}}$ and a method for selecting one, called a *fit* or *fitted model* $\hat{f}_{\mathcal{T}}$, given training data $\mathcal{T}$. Many models fall into the framework of *empirical risk minimization (ERM)*. First, we select a *loss function*

$$\mathcal{L}_f(X, Y) = d(f(X), Y), \tag{3}$$

which weights the discrepancy between Y and our prediction $f(X)$ by some penalty function $d : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$.

A choice of loss induces an *empirical risk* for candidate functions f, which is simply the average loss per data point:

$$\mathcal{R}_{\mathcal{T}}(f) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_f(x_i, y_i). \tag{4}$$

The ERM principle is to choose the function $f \in \mathcal{F}$ that minimizes empirical risk:

$$\hat{f}_{\mathcal{T}} = \text{argmin}_{f\in\mathcal{F}}\mathcal{R}_{\mathcal{T}}(f). \tag{5}$$

The *training error* is the minimal empirical risk itself:

$$\mathcal{E}_{\mathbf{x},\mathbf{y}} = \mathcal{R}_{\mathcal{T}}(\hat{f}_{\mathcal{T}}), \tag{6}$$

where the subscript indicates dependence on both inputs $\mathbf{x}$ and outputs $\mathbf{y}$. Since we have chosen the function to minimize risk, it suffers from Goodhart's curse:

> When a measure becomes a target, it ceases to be a good measure.

Our goal is to assess models using quantities that are non-Goodhartian and which we can actually calculate.

II. OPTIMISM

Suppose we have a sequence of models $\mathcal{F}^{(a)}$ with fits $\hat{f}^{(a)}_{\mathcal{T}}$ and want to choose the best. Naively, you might guess that comparing training errors $\mathcal{E}^{(a)}_{\mathcal{T}} = \mathcal{R}_{\mathcal{T}}(\hat{f}^{(a)}_{\mathcal{T}})$ will tell us which is "best". But as per Goodhart's curse, model can overfit and do badly on unseen data, with a training error unrelated to extra-sample performance. Such a model does not *generalize*.

The most direct way to measure this failure to generalize is to consider the expected loss on the whole domain $\mathcal{X} \times \mathcal{Y}$, also called *expected risk*:

$$\mathcal{E}(f) = \mathbb{E}_{(X,Y)}\left[\mathcal{L}_f(X, Y)\right], \tag{7}$$

using the joint distribution $\mathbb{P}(X, Y) = \mathbb{P}(X)\mathbb{P}(Y|X)$ as described above. There is no subscript on the LHS of (7) since we are not conditioning on any observed values.

The problem with (7) is that it is *global*; it uses inputs we have not seen, and a global probability distribution $\mathbb{P}(X, Y)$ we don't know. We have no hope of calculating it! However, there is a quantity less susceptible to overfit than training error $\mathcal{E}_{\mathbf{x},\mathbf{y}}$, and less global than expected risk $\mathcal{E}$. Our inspired notational guess is $\mathcal{E}_{\mathbf{x}}$. Let's see what this notation could mean!

If $\mathcal{E}_{\mathbf{x},\mathbf{y}}$ conditions on both inputs and outputs, $\mathcal{E}_{\mathbf{x}}$ conditions only on inputs. It will average over new outputs Y_i at each of the fixed x_i, with conditional distribution $\mathbb{P}(Y_i|X = x_i)$. Thus, we define the *in-sample* or simply *sample error* as

$$\mathcal{E}_{\mathbf{x}}(f) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_{Y_i}\left[\mathcal{L}_f(x_i, Y_i)\right]. \tag{8}$$

Writing $\mathcal{E}_{\mathbf{x}}(\hat{f}_{\mathbf{x},\mathbf{y}})$ is a pedantic but clear way to refer to the fact that we fit $\hat{f}$ on the training set $\mathcal{T} = (\mathbf{x}, \mathbf{y})$, then fix the x_i and average over conditioned random variables Y_i.

Optimism is the gap between sample and training error:

$$\text{op}_{\mathbf{x},\mathbf{y}} = \mathcal{E}_{\mathbf{x}}(\hat{f}_{\mathbf{x},\mathbf{y}}) - \mathcal{E}_{\mathbf{x},\mathbf{y}}. \tag{9}$$

It's called this because $\mathcal{E}_{\mathbf{x},\mathbf{y}}$ is usually optimistic due to overfit. This is still not an estimable quantity, since we don't know $\mathbb{P}(Y|X)$, and usually don't have enough data (multiple observations per x_i) to Monte Carlo estimate. What can we do?

In statistical modeling as in statistical mechanics, it is easier to understand average behaviour than specific trajectories. Here, the fixed response vector $\mathbf{y}$ is used to train $\hat{f}_{\mathbf{x},\mathbf{y}}$, and also enters the empirical risk. If our goal is model selection, and not assessing the performance of a specific $\hat{f}$, we can average over $\mathbf{y}$, to obtain the *expected optimism*:

$$\omega_{\mathbf{x}} = \mathbb{E}_{\mathbf{y}}[\text{op}_{\mathbf{x},\mathbf{y}}]. \tag{10}$$

To be clear, for each value of $\mathbf{y}$, we fit a new $\hat{f}_{\mathbf{x},\mathbf{y}}$. It will turn out that expected optimism *can* be estimated.

III. REGRESSION AND COVARIANCE

Let's use the *squared error loss* $\mathcal{L}_f(X, Y) = (f(X) - Y)^2$. When performing regression, the theoretically optimal prediction f^* is the *conditional mean*:

$$f^*(X) = \text{argmin}_c \mathbb{E}\left[(Y-c)^2|X\right] = \mathbb{E}_Y[Y|X]. \tag{11}$$

More generally, it is a conditional minimum of the loss called the *regression function* or *regressor*:

$$f^*_{\mathcal{L}}(X) = \text{argmin}_c \mathbb{E}\left[\mathcal{L}(c, Y)|X\right].$$

We want our model to learn this regressor from training data, and not just the response values y_i we happen to feed it.

For a fitted model $\hat{f}_{\mathbf{x},\mathbf{y}}$, let $\hat{y}_{\mathbf{y},i} = \hat{f}_{\mathbf{x},\mathbf{y}}(x_i)$ be the predicted response. One way to quantify how well we are doing at learning the regressor $f^*_{\mathcal{L}}$ is to see how much specific responses y_i "pull" the prediction $\hat{y}_i$ away from the unknown, but y_i-independent, regressor prediction $f^*_{\mathcal{L}}(x_i)$. Mathematically, we can view overfitting as a *correlation* between training values y_i and the induced prediction $\hat{y}_i$ in the fit.

Expected optimism measures exactly this correlation for a wide range of loss functions. Let's consider squared loss for concreteness. We start by rewriting the training error averaged over $\mathbf{y}$, sometimes indicated with an overline for compactness. The term at input x_i is of the form:

$$\begin{aligned}\mathbb{E}_{\mathbf{y}}\left[(\hat{y}_{\mathbf{y},i} - y_i)^2\right] &= \mathbb{E}_{\mathbf{y}}\left[\left(\hat{y}_{\mathbf{y},i} - \overline{\hat{y}_{\mathbf{y}',i}} + \overline{\hat{y}_{\mathbf{y}',i}} - y_i\right)^2\right] \\ &= \text{var}_{\mathbf{y}}[\hat{y}_{\mathbf{y},i}] + \mathbb{E}_{\mathbf{y}}\left[\left(\overline{\hat{y}_{\mathbf{y}',i}} - y_i\right)^2\right] \\ &\quad - 2\,\text{cov}(\hat{y}_{\mathbf{y},i}, y_i),\end{aligned}$$

where the cross-term is the covariance, defined by

$$\text{cov}(X, Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right].$$

The second term can be further split into bias plus variance, but we won't need to do that here.

We can perform a similar decomposition for sample error, replacing y_i by the random variable Y_i and taking an expectation $\mathbb{E}_Y$. The only difference is that the covariance disappears,

$$\mathbb{E}_{\mathbf{y}}\mathbb{E}_Y\left[\left(\hat{y}_{\mathbf{y},i} - \overline{\hat{y}_{\mathbf{y}',i}}\right)\left(\overline{\hat{y}_{\mathbf{y}',i}} - Y\right)\right] = 0,$$

since the first term is independent of Y, so it factors out and vanishes on average. Similarly, the middle terms are identical since they perform the same average over y_i.

Since the expected optimism (10) is the difference between expected sample error and training error, everything cancels except the covariance:

$$\omega_{\mathbf{x}} = \frac{2}{N}\sum_{i=1}^{N}\text{cov}(\hat{y}_{\mathbf{y},i}, y_i). \tag{12}$$

Thus, as claimed, expected optimism quantifies the degree of correlation between response (y_i) and the induced fit ($\hat{y}_{\mathbf{y},i}$).

IV. SMOOTH OPERATOR

We'll end by calculating covariance and hence average optimism in the large class of *linear smoother* models, including linear regression and smoothing splines. They are defined by the existence of an $N \times N$ *smoother matrix* $\mathbf{S}$, depending only on $\mathbf{x}$, which maps $\mathbf{y}$ to the fitted values $\hat{\mathbf{y}}$:

$$\hat{\mathbf{y}} = \mathbf{S}_{\mathbf{x}}\mathbf{y}. \tag{13}$$

The covariance between the response and fit at x_i is then

$$\begin{aligned}\text{cov}(\hat{y}_i, y_i) &= \mathbb{E}\left[(\hat{\mathbf{y}} - \mathbb{E}[\hat{\mathbf{y}}])(\mathbf{y} - \mathbb{E}[\mathbf{y}])^T\right]_{ii} \\ &= \mathbf{S}_{\mathbf{x}} \cdot \mathbb{E}\left[(\mathbf{y} - \mathbb{E}[\mathbf{y}])(\mathbf{y} - \mathbb{E}[\mathbf{y}])^T\right]_{ii} \\ &= \sum_{j=1}^{N}(\mathbf{S}_{\mathbf{x}})_{ij}\mathbb{E}[(y_j - \mathbb{E}[y_j])(y_i - \mathbb{E}[y_i])].\end{aligned}$$

Since we assume the y_i are independent, this cross-term is diagonal, giving $\delta_{ij}\text{var}[y_i]$. Thus,

$$\text{cov}(\hat{y}_i, y_i) = (\mathbf{S}_{\mathbf{x}})_{ii}\text{var}[y_i]. \tag{14}$$

If we assume, further, that the variance $\text{var}[y_i] = \sigma^2$ is independent of x_i, we obtain

$$\omega_{\mathbf{x}} = \frac{2\sigma^2}{N}\text{Tr}[\mathbf{S}_{\mathbf{x}}]. \tag{15}$$

The trace of $\mathbf{S}_{\mathbf{x}}$ can be identified with the number of *degrees of freedom* of the model; thus, overfit is related to (a) model complexity $\text{Tr}[\mathbf{S}_{\mathbf{x}}]$, (b) underlying noise σ^2, and (c) the size of the training set N. Since noisier outputs y_i pull the fit further away from the regressor prediction $f^*(x_i)$, this makes sense! I'm not sure if 2 has a nice interpretation.

V. MODEL SELECTION

To compare models $\mathcal{F}^{(a)}$, we would like to find the generalization error $\mathcal{E}^{(a)}$ for each. We can't, so we settle for the sample error $\mathcal{E}^{(a)}_{\mathbf{x}}$, which is training error $\mathcal{E}^{(a)}_{\mathbf{x},\mathbf{y}}$ plus optimism $\text{op}^{(a)}_{\mathbf{x},\mathbf{y}}$. Sadly, we don't know optimism for specific $\mathbf{y}$, but the average $\omega^{(a)}_{\mathbf{x}}$ is a property of our model we can sometimes calculate. Great! Except that, now we don't know the *average* training error.

The compromise is to take the actual training error as an *estimator* for the average. Suppose we're comparing different linear smoothers with complexity $d^{(a)} = \text{Tr}[\mathbf{S}^{(a)}]$. In practice, we also need to estimate the underlying variance as $\tilde{\sigma}^2$. Thus, our model selection criterion is to minimize

$$C^{(a)}_p = \tilde{\mathcal{E}}^{(a)}_{\mathbf{x}} = \mathcal{E}^{(a)}_{\mathbf{x},\mathbf{y}} + \frac{2d^{(a)}\tilde{\sigma}^2}{N}. \tag{16}$$

This is a "statistic", i.e. an estimator of the average sample error, called *Mallow's C_p statistic*, related to the *Aikeke information criterion (AIC)*. This estimator is asymptotically *biased* (can be wrong with infinite samples) but *efficient* (minimizes prediction error). Let's leave it there!